

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.10 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 Let z_i , $i = 1, 2, 3, 4, 5, 6$ be the roots of equation $z^6 + z^4 = 2$, then $\sum_{i=1}^6 |z_i|^4$ is equal to
(A) 4 (B) 6 (C) 8 (D) 10
- Q.2 If m and M denotes the minimum and maximum value of $|2z + 1|$ where $|z - 2i| \leq 1$ then $(m + M)^2$ is equal to
(A) 17 (B) 34 (C) 51 (D) 68
- Q.3 Number of solutions of the equation $z^3 + \frac{3(\bar{z})^2}{|z|} = 0$, where $z \in \mathbb{C}$ is equal to
(A) 2 (B) 3 (C) 5 (D) 6
- Q.4 If $|z_1 + iz_2| = |z_1| + |z_2|$ and $|z_1| = 3$ and $|z_2| = 4$, then area of triangle ABC, if affix of A, B and C are z_1, z_2 and $\frac{z_2 - iz_1}{1 - i}$ respectively, is equal to
(A) $\frac{5}{2}$ (B) 0 (C) $\frac{25}{2}$ (D) $\frac{25}{4}$
- Q.5 If $|z_1| = 2$ and $(1 - i)z_2 + (1 + i)\bar{z}_2 = 8\sqrt{2}$, then the minimum value of $|z_1 - z_2|$ equals
(A) 4 (B) 3 (C) 2 (D) 1
- Q.6 Image of the point whose affix is $\frac{2-i}{3+i}$, in the line $(1+i)z + (1-i)\bar{z} = 0$ is the point whose affix is
(A) $\frac{1+i}{2}$ (B) $\frac{1-i}{2}$ (C) $\frac{-1+i}{2}$ (D) $\frac{-1-i}{2}$
- Q.7 If z_1, z_2, z_3 and z_4 are the complex roots of the equation $z^4 + 3z^2 + 1 = 0$, then the value of $\prod_{i=1}^4 (4 + z_i^2)$ is
(A) 9 (B) 12 (C) 25 (D) 27

Q.8 If $|z - 1 - 2i| = r$, $0 < r < 5$ and a and b be the corresponding complex numbers z for which $|z - 5 + i|$ is minimum and maximum respectively, then $\arg\left(\frac{a - 5 + i}{b - 1 - 2i}\right)$ is equal to

- (A) 0 (B) $\frac{\pi}{2}$ (C) π (D) $\frac{\pi}{6}$

Q.9 If z is a complex number lying in the fourth quadrant of argand plane and $\left|\frac{kz}{k+1} + 2i\right| > \sqrt{2}$ is true for all real values of k ($k \neq -1$), then range of $\arg(z)$ is

- (A) $\left(-\frac{\pi}{8}, 0\right)$ (B) $\left(-\frac{\pi}{6}, 0\right)$ (C) $\left(-\frac{\pi}{4}, 0\right)$ (D) $\left(-\frac{\pi}{3}, 0\right)$

Q.10 If $\arg\left(\frac{z_1 - \frac{z}{|z|}}{\frac{z}{|z|}}\right) = \frac{\pi}{2}$ and $\left|\frac{z}{|z|} - z_1\right| = 3$, then $|z_1|$ is equal to

- (A) $\sqrt{26}$ (B) $\sqrt{10}$ (C) $\sqrt{3}$ (D) $2\sqrt{2}$

[PARAGRAPH TYPE]

Q.11 to Q.15 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 11 to 13

Let z and ω be two complex numbers satisfying $z + \bar{\omega} = z^2$ and $\omega + \bar{z} = \omega^2$.

Q.11 Number of ordered pairs (z, ω) is equal to

- (A) 1 (B) 2 (C) 3 (D) 4

Q.12 Let z_r and ω_r , $r = 1, 2, 3, \dots, n$ be the solutions of the equations, then $\sum z_r^2 + \sum \omega_r^2$ is equal to

- (A) 4 (B) 6 (C) 8 (D) 10

Q.13 The sum of an infinite terms of G.P. $|z| + \frac{|z|}{|\omega+1|} + \frac{|z|}{|\omega+1|^2} + \frac{|z|}{|\omega+1|^3} + \dots \infty$ is equal to

- (A) 1 (B) 3 (C) 5 (D) not possible

Paragraph for question nos. 14 & 15

Let $z_1 = 3 + i0$ and $z_2 = 7 + i0$ ($i = \sqrt{-1}$) represents two points M and N respectively on complex plane. Let the curve C_1 be the locus of point P(z) satisfying $|z - z_1|^2 + |z - z_2|^2 = 10$ and the curve C_2 be the locus of point Q(z) satisfying $|z - z_1|^2 + |z - z_2|^2 = 16$.

Q.14 The locus of point from which tangents drawn to C_1 and C_2 are perpendicular is

- (A) $|z - 3| = 2$ (B) $|z - 5| = \sqrt{5}$ (C) $|z - 5| = 3$ (D) $|z - 5| = 4$

Q.15 The least distance between two curves C_1 and C_2 is

- (A) 1 (B) 2 (C) 3 (D) 4

[INTEGER TYPE]

Q.16 to Q.20 are "Integer Type" questions. (The answer to each of the questions are upto 4 digits)

Q.16 Let ω ($\omega \neq 1$) be a cube root of unity such that $(1 + \omega^2)^8 = a + b\omega$ where $a, b \in \mathbb{R}$, then find the value of $|2a^{-1} + 2b^{-1}|$.

Q.17 Let $z_1, z_2, z_3 \in \mathbb{C}$ such that $|z_1| = |z_2| = |z_3| = 1$. If $z_1 + z_2 + z_3 \neq 0$ and $z_1^2 + z_2^2 + z_3^2 = 0$, then find the value of $|z_1 + z_2 + z_3|$.

Q.18 Let $A(z_1)$ and $B(z_2)$ be lying on the curve $z - 3 - 4i = \frac{25}{\bar{z} - 3 + 4i}$ where $|z_1|$ is maximum. Now, $A(z_1)$ is rotated about the origin in anticlockwise direction through 90° reaching at $P(z_0)$. If A, B and P are collinear then find the value of $(|z_0 - z_1| \cdot |z_0 - z_2|)$.

Q.19 If a, b and c are different odd integers, then find the minimum value of $\frac{1}{2} |a + b\omega + c\omega^2|^2$ where ω is an imaginary cube root of unity.

Q.20 Let C be the set of complex numbers and $z \in \mathbb{C}$. If $S = \{z : |z + \bar{z}| + 2|z - \bar{z}| = 4 \text{ and } |z| \text{ is minimum}\}$ then area of the polygon formed by all the points in S taking as a vertices is $\frac{p}{q}$ where $p, q \in \mathbb{N}$. Find the least value of $(p - q)$.

ANSWER KEY

Q.1	D	Q.2	D	Q.3	C	Q.4	D	Q.5	C
Q.6	C	Q.7	C	Q.8	A	Q.9	C	Q.10	B
Q.11	B	Q.12	C	Q.13	B	Q.14	B	Q.15	A
Q.16	[0004]	Q.17	[0002]	Q.18	[0100]	Q.19	[0006]	Q.20	[7]

